Assignment 2

Q1. Lexical Ambiguity

Given Sentence:

“The bank will not open until the river recedes.”

a) Identify and explain the type of lexical ambiguity present.

The word “bank” creates lexical ambiguity in this sentence because it can have more than one meaning.

For example, “bank” can refer to:

- A financial institution, where people deposit or withdraw money.
- The land beside a river, also called a river bank.

In the given sentence, the words “river” and “recedes” make it clear that “bank” refers to the land beside the river and not a financial institution.

Therefore, the ambiguity occurs because the same word has multiple possible meanings, and the correct meaning has to be understood from the context.

b) How can WordNet or BabelNet help resolve this ambiguity?

WordNet organizes words according to their different meanings, known as synsets. For the word “bank,” WordNet provides different senses, including the financial meaning and the river-side meaning.

An NLP system can look at the words surrounding “bank.” In this sentence, words such as “river” and “recedes” are strongly related to the river-bank meaning. Therefore, the system can select that sense instead of the financial meaning.

BabelNet can also help by connecting words and concepts across languages and showing semantic relationships between them. It can help the system understand that “bank” in this context is related to a river-side landform.

Thus, WordNet and BabelNet provide the possible meanings, while the surrounding context helps the NLP system choose the most suitable one.

c) Why might automatic systems still fail in some contexts?

Automatic systems may still make mistakes when the context is short, unclear, or does not contain enough information to identify the intended meaning.

For example, if a sentence only says:

“I went to the bank.”

it is difficult to know whether the person went to a financial institution or to a river bank.

Another problem is that some situations require common-sense or real-world knowledge. Simply looking at individual keywords may not always be enough to understand the intended meaning.

Therefore, even when resources such as WordNet and BabelNet are available, automatic ambiguity resolution cannot always be completely accurate.

Q2. Word Sense Disambiguation Using Naïve Bayes

A company wants to build an NLP system that can classify the correct meaning of a word using Naïve Bayes.

a) How can Naïve Bayes be applied to Word Sense Disambiguation (WSD)?

Word Sense Disambiguation (WSD) is the process of identifying the correct meaning of a word based on the context in which it appears.

Naïve Bayes can treat each possible meaning of a word as a separate class. The system is first trained using examples where the correct meaning is already known.

For example, consider the word “bank.”

- “I deposited money in the bank.” → Financial bank
- “They sat on the river bank.” → River bank

When a new sentence is given, the system examines the surrounding words and calculates the probability of each possible meaning.

The basic idea can be represented as:

$P(\text{Sense} \mid \text{Context}) \propto P(\text{Sense}) \times P(\text{Context} \mid \text{Sense})$

The sense with the highest probability is selected as the predicted meaning.

The “naïve” assumption is that the context features are conditionally independent of each other when the sense is known. This assumption is not completely true for natural language, but it makes the model simple and efficient.

b) What are the key limitations of Naïve Bayes compared to Lesk's knowledge-based approach?

One major limitation of Naïve Bayes is that it is a supervised learning method, so it needs labelled training data. Creating a large dataset where every occurrence of a word is labelled with its correct sense can take a lot of time and effort.

Another limitation is its assumption that features are independent. In natural language, words are often strongly connected to each other, so this assumption may not always represent the actual relationship between words.

Naïve Bayes can also perform poorly when a particular word sense is rare or when the test sentence contains words that were not well represented in the training data.

In comparison, Lesk's algorithm is a knowledge-based approach. It uses dictionary definitions or glosses and compares them with the context. The sense whose definition has the greatest overlap with the context is selected.

The main difference can be summarized as follows:

Naïve Bayes	Lesk
Supervised, data-driven approach	Knowledge-based approach
Requires labelled training data	Uses dictionary definitions
Learns probabilities from examples	Compares context with word definitions
Assumes feature independence	Mainly depends on word overlap
Can perform well with sufficient training data	Can work without a large labelled dataset

However, Lesk also has limitations. A simple word-overlap approach may fail when two meanings are semantically related but use different words in their definitions.

c) Suggest one improvement using a semi-supervised method such as Yarowsky's algorithm.

A useful improvement would be to use Yarowsky's semi-supervised algorithm.

Instead of manually labelling a very large dataset, the system can begin with a small number of labelled examples for each word sense. It can then use these examples to train an initial classifier.

The process can be carried out as follows:

1. Start with a small set of labelled examples for each word sense.
2. Train an initial classifier using these examples.
3. Apply the classifier to a large collection of unlabelled sentences.
4. Select the predictions for which the system has high confidence.
5. Add these confidently classified examples to the training dataset.
6. Retrain the classifier using the larger dataset.

7. Repeat the process until enough reliable training examples are obtained.

For example, if the system confidently identifies several unlabelled sentences containing “bank” as referring to a river bank, those sentences can be added to the training examples for that particular sense.

This approach reduces the amount of manual labelling required and allows the system to learn from a much larger amount of unlabelled text.

Q3. Anaphora Resolution and Hobbs' Algorithm

Given Sentence:

“John met Alex after he finished the project.”

a) Identify the anaphora and explain why reference resolution is difficult here.

The word “he” is the anaphoric expression in the sentence.

The problem is that “he” could refer to either John or Alex, since both are male persons mentioned earlier in the sentence.

There are therefore two possible interpretations:

- Interpretation 1: John finished the project.
- Interpretation 2: Alex finished the project.

The sentence itself does not provide enough information to clearly identify which person “he” refers to.

This is known as anaphora resolution, where an NLP system has to determine which previously mentioned entity a pronoun refers to.

b) How would Hobbs' algorithm attempt to resolve “he”?

Hobbs' algorithm is a syntactic approach used for pronoun or anaphora resolution. It makes use of the syntactic parse tree of a sentence to find possible antecedents.

For the sentence:

“John met Alex after he finished the project.”

the system first identifies “he” as the pronoun that needs to be resolved.

It then examines the syntactic structure and considers previously mentioned noun phrases such as:

- John

- Alex

The algorithm checks these candidates based on their position and syntactic relationship with the pronoun. Candidates that do not satisfy the required syntactic conditions can be eliminated.

However, in this particular sentence, syntax alone may not provide a definite answer because both John and Alex can potentially satisfy the basic requirements.

Therefore, additional context or real-world knowledge may be required to determine the intended antecedent.

c) Give a real-world scenario where syntactic constraints alone fail to ensure coherence.

Consider the following customer-support conversation:

“The customer spoke to the manager after he received the complaint.”

Here, the pronoun “he” could refer to either the customer or the manager.

Suppose the earlier conversation explains that the manager received the complaint. A human reader can use this additional information to understand that “he” refers to the manager.

This shows that syntactic rules alone are sometimes not enough. In real-world conversations, news articles, emails, and workplace communication, an NLP system may need information from previous sentences, real-world knowledge, and the overall context to correctly resolve a reference.

Therefore, syntactic constraints are useful for reducing the number of possible interpretations, but they cannot always guarantee correct discourse understanding.